

Dermatologic Pharmacology

By

Yaschilal Muche (B.Pharm, MSc)

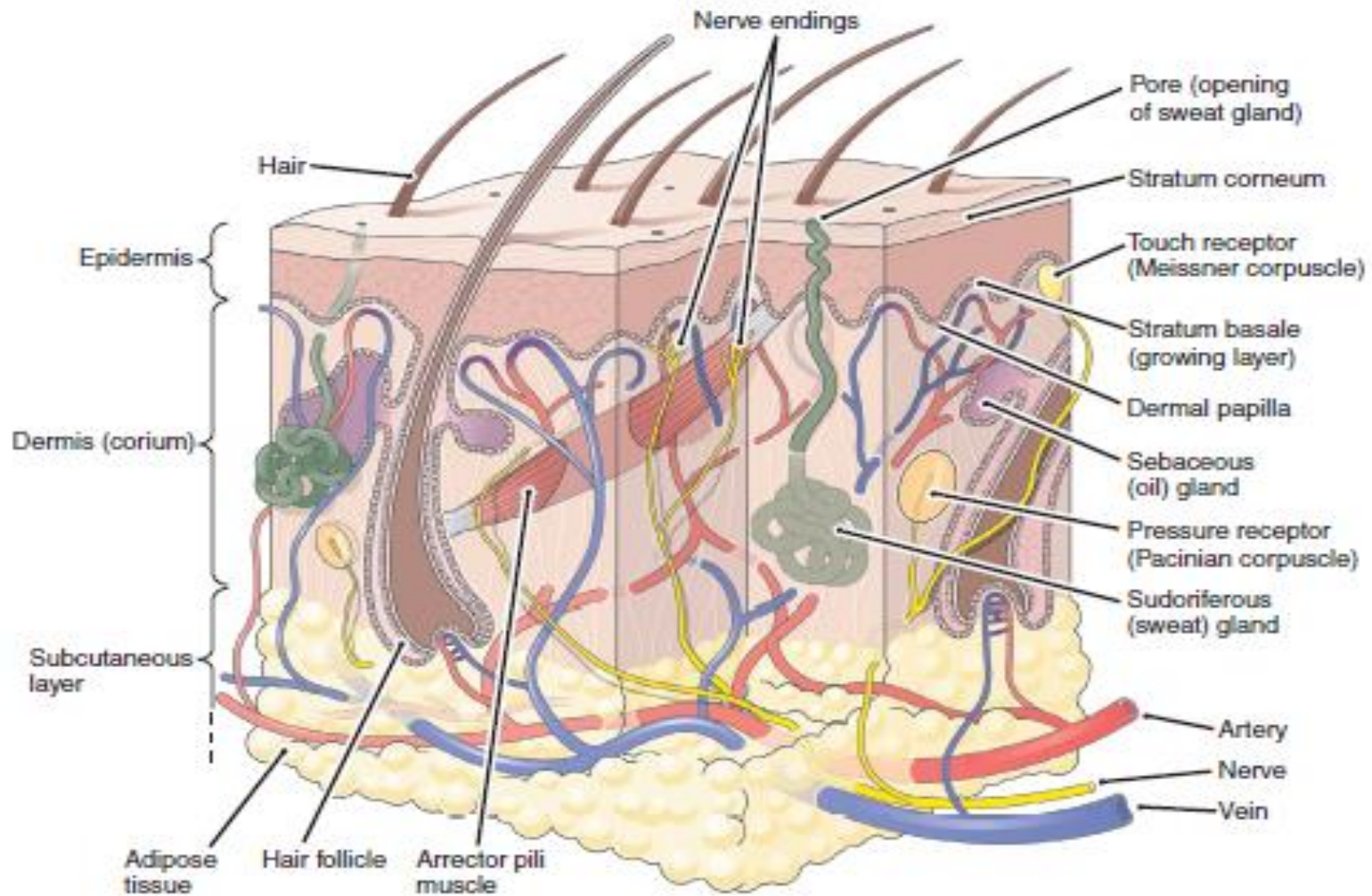
Assistant Professor of Pharmacology, WU

2024

Introduction

- The skin is composed of two main layers, the epidermis and the dermis
 - The outermost layer of the epidermis, the stratum corneum, consists of **a lipophilic keratinous** layer of skin.
- Use of topical agents for treatment of dermatologic disorders is
 - convenient
 - minimizes systemic adverse effects.
- Topical agents may be formulated as
 - sprays, powders, lotions, creams, pastes, packed dressings, ointments, and aerated foams.

Cross section of the skin;



- The general pharmacokinetic principles governing the use of drugs applied to the skin are
 - the same as those involved in other ROA
- Major variables that determine pharmacologic response to drugs applied to the skin include the following:
 - 1. Regional variation in drug penetration:**
 - For example, the scrotum, face, axilla, and scalp are far more permeable than the forearm and may require less drug for equivalent effect.

2. Concentration gradient:

- Increasing the conc/n gradient increases the mass of drug transferred per unit time
 - Thus, resistance to topical corticosteroids can sometimes be overcome by use of higher concentrations of drug.

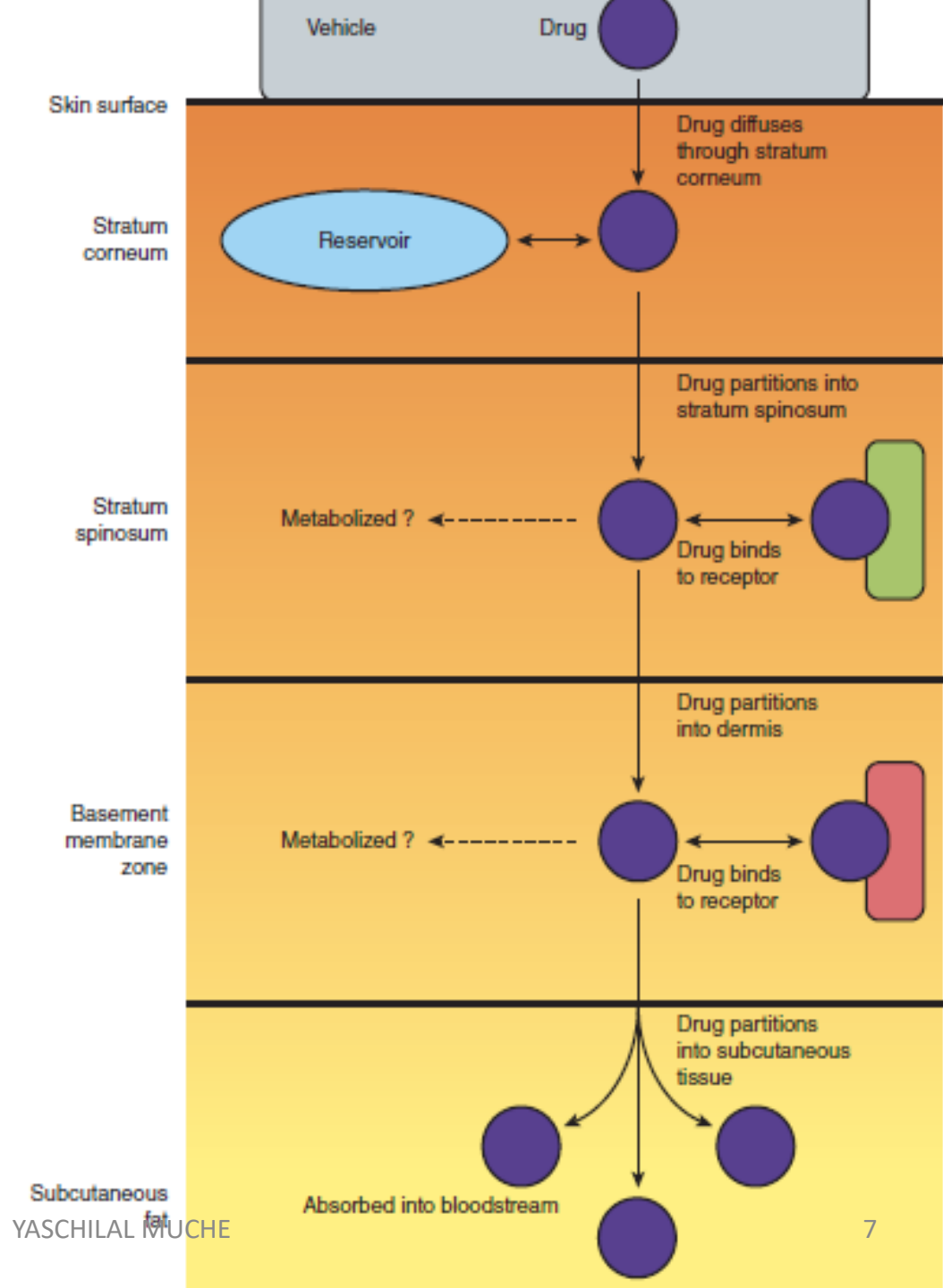
3. Dosing schedule:

- Because of its physical properties, skin acts as a reservoir for many drugs.
 - As a result, the “local half-life” may be long enough to permit once-daily application of drugs with short systemic half-lives.

4. Vehicles and occlusion:

- An appropriate vehicle
 - maximizes the ability of the drug to penetrate the outer layers of the skin.
- Occlusion (application of a plastic wrap to hold the drug and its vehicle in close contact with the skin) is
 - extremely effective in maximizing efficacy.

FIG; Schematic diagram of percutaneous absorption



Agents for acne vulgaris

- Acne vulgaris is a common skin disorder characterized by
 - pimples, comedones, pustules, and sometimes nodules and scarring.
- Acne occurs due to alterations in pilosebaceous units-
 - skin structures that contain a hair follicle and a sebaceous gland.
- Androgens stimulate sebaceous glands, thereby producing sebum that leads to follicular keratinization and obstruction.
- Propionibacterium acnes, part of the normal skin flora, can enter the clogged pore and multiply, causing
 - redness and inflammation and leading to papillary, pustular, and cystic acne.
- Treatments for acne
 - **reduce sebum production or control P. acnes.**

Retinoids

- Retinoids are derivatives of vitamin A
- are highly effective in the t/t of acne, as well as other skin conditions such as
 - psoriasis and photoaging.
- **Tretinoin** and **isotretinoin** are first-generation retinoids that are used for the management of acne.
- Third-generation retinoids; **adapalene** and **tazarotene**.
 - are less irritating and more effective than 1st generation retinoids
 - are first-line therapy for **comedonal and inflammatory** acne.

- Retinoids are applied topically, with the exception of *isotretinoin* (PO).
- *Due to* the adverse effect profile, use of *isotretinoin*
 - should be reserved for severe cystic acne.
- MOA; are comedolytic and anti-inflammatory
 - Unlike the third-generation agents, the first-generation agents influence sebum production.
- AEs; Irritation, dryness, skin peeling, Photosensitivity, dry mucous membranes and dry eyes
 - Suicide or suicide attempts (oral *isotretinoin*)

- Retinoids are contraindicated in pregnancy.

Benzoyl peroxide

- *Benzoyl peroxide* is considered the first-line agent for mild to moderate acne
 - with **no inflammation**.
- MOA;
 - antiseptic effects against *P. acnes* as well as opening of the pores.
- is a topical agent that is available in many OTC acne treatment products, as well as some prescription products.
- Local AEs; Dry skin, peeling, and irritation.

Salicylic acid

- Topical *salicylic acid*, penetrates the pilosebaceous unit and works as an **exfoliant** to clear comedones.
 - Its comedolytic effects are not as pronounced as those of the retinoids.
- The drug has mild anti-inflammatory activity and is keratolytic at higher concentrations.
- *Salicylic acid* is used as a treatment for mild acne and is available in many OTC facial washes and medicated treatment pads.
- AEs; Mild skin peeling, dryness, and local irritation.

Azelaic acid

- Is a dicarboxylic acid, has antibacterial activity against *P. acnes* as well as anti-inflammatory actions.
- Azelaic acid normalizes keratinization and
 - is anticomedogenic.
- It is available as a topical preparation for the treatment of **mild to moderate inflammatory acne**.
- **AE**; well tolerated, with mild skin irritation

Antibiotics

- *P. acnes* is a gram-positive rod that is associated with
 - inflammatory lesions in acne.
- For moderate to severe acne with inflammatory lesions, use of topical or oral antibiotics is useful for inhibition of *P. acnes*.
- Topical formulations of ***erythromycin*** and ***clindamycin*** (preferred)
 - may be combined with benzoyl peroxide or the retinoids for better effectiveness.
- *Dapsone* is available as a topical gel for acne (unknown mechanism)

- Oral antibiotics commonly used for the management of moderate to severe acne include
 - **minocycline, doxycycline**, and erythromycin.
 - *Erythromycin* is used infrequently due to GI adverse effects.

DRUGS FOR PSORIASIS



Fig: Psoriasis. A large, scaly, erythematous plaque.

- Psoriasis is a skin disease that presents with erythematous scaling plaques.
- It manifests with increased epidermal cell proliferation.
- Psoriasis appears to have both genetic factors and T-cell-mediated immune components.

- The majority of patients have mild to moderate psoriasis, and
 - this can be managed with **topical** treatments including retinoids, vitamin D analogues, keratolytic agents, and corticosteroids.
- More severe cases require systemic therapy with
 - phototherapy (*methoxsalen followed by UVA or UVB alone*),
 - *methotrexate*,
 - *cyclosporine*, or
 - *biologic response modifiers (for example, etanercept, adalimumab)*.

A. Retinoids

- *Tazarotene is a topical retinoid used for the treatment of plaque psoriasis.*
 - AEs are similar to other retinoids.
- *Acitretin is a second-generation retinoid used orally in the treatment of pustular forms of psoriasis.*
 - Since ingestion of ethanol can increase transesterification of **acitretin to etretinate**, *ethanol is contraindicated with this agent.*
 - AEs; Cheilitis, pruritus, peeling skin, and hyperlipidemia

B. Vitamin D analogues

- *Calcipotriene* and *calcitriol* are synthetic vitamin D3 derivatives used topically to treat plaque psoriasis.
 - They inhibit keratinocyte proliferation and increase keratinocyte differentiation.
 - Their therapeutic effectiveness does not appear to decrease upon continued use.
 - Transient elevations in calcium levels have been reported in some patients.
- AEs; itching, dryness, burning irritation, and erythema.

C. Keratolytic agents

- Keratolytic agents such as *coal tar* and *salicylic acid*
 - *are effective in* localized psoriasis, especially on the scalp.
- They improve corticosteroid penetration.
- *Coal tar* inhibits
 - excessive skin cell proliferation and may also have anti-inflammatory effects.

AGENTS FOR PIGMENTATION DISORDERS



- **hydroquinone** *and* **methoxsalen**, which are used for the t/t of **hyperpigmented** skin conditions and **vitiligo**, respectively

Fig; The palm is frequently affected by vitiligo.

A. Hydroquinone

- is a topical skin whitening agent that reduces
 - hyperpigmentation associated with freckles and melasma.
- It is often used in combination with topical retinoids
 - to treat the signs of photoaging.
- MOA;
 - *inhibition of* the tyrosinase enzyme required for melanin synthesis.
- Hydroquinone lightens the skin temporarily and is commonly used as a 4% preparation.

B. Methoxsalen, Trioxsalen

- *Methoxsalen* is a photoactive substance (psoralen) that stimulates melanocytes and
 - is used as a repigmentation agent for patients with vitiligo.
- It must be photoactivated by UV radiation to form a DNA adduct inhibiting DNA replication by a method called PUVA (psoralen plus UVA radiation).
- *Methoxsalen* inhibits cell proliferation and promotes cell differentiation of epithelial cells.
- Topical *methoxsalen* may be used for small patches of vitiligo, and oral therapy is used for more widespread disease.
- Because of the possibilities for aging of the skin and possible carcinogenicity, it is used with caution.

TRICHOGENIC AGENTS

- *Minoxidil and finasteride are trichogenic agents that are indicated for the treatment of*
 - androgenic alopecia (“male pattern baldness”).
- MOA:
 - *Minoxidil*; not fully known, it is believed to act, at least in part, by shortening the rest phase of the hair cycle.
 - *Finasteride*; decreases scalp and serum DHT concentrations, *thus inhibiting* a key factor in the etiology of androgenic alopecia.

TOPICAL ANTIBACTERIAL AGENTS

- Organisms such as staphylococci and streptococci can cause
 - folliculitis, abscesses, cellulitis, impetigo, and many pus-forming infections.
- Several gram-positive and gram-negative bacteria cause infections that
 - are not limited to the skin and may cause serious systemic diseases



Fig; Impetigo on the face

A. Gram-positive infections

- *Bacitracin* is a peptide antibiotic active against many
 - gram-positive organisms.
- It is used mainly in topical formulations;
 - if used systemically, it is toxic.
- *Bacitracin* is mostly used for the prevention of
 - skin infections after burns or minor scrapes.
- It is frequently found in combination products with neomycin and/or polymyxin.

- **Mupirocin** *is a protein synthesis inhibitor* that is useful in treating
 - impetigo (a contagious skin infection caused by streptococci or staphylococci) and
 - other serious gram positive skin infections, including infections caused by *MRSA*.
- **Retapamulin** is a newer protein synthesis inhibitor that treats impetigo.

B. Gram-negative infections

- *Polymyxin B* is a cyclic hydrophobic peptide that disrupts
 - the bacterial cell membrane of gram-negative organisms.
- commonly combined with *neomycin and bacitracin* (“triple antibiotic”)
 - for the prevention of skin infections after minor skin trauma.
- *Neomycin* in combination with other agents and also *gentamicin can be used*
 - to treat skin infections caused by gram-negative organisms such as *Pseudomonas*, *E. coli*, and *Klebsiella* sp.

Summary of agents for selected dermatologic disorders:

ACNE VULGARIS AGENTS

Adapalene DIFFERIN
Azelaic acid AZELEX
Benzoyl peroxide VARIOUS
Isotretinoin VARIOUS
Tazarotene TAZORAC
Tretinoin RETIN-A

TOPICAL ANTIBACTERIAL AGENTS

Bacitracin
Clindamycin CLEOCIN T
Dapsone ACZONE
Erythromycin BENZAMYCIN
Gentamicin
Mupirocin BACTROBAN
Neomycin (with *bacitracin* and *polymyxin B*) TRIPLE ANTIBIOTIC
Retapamulin ALTABAX

ECTOPARASITICIDES

Crotamiton EURAX
Lindane
Permethrin ELIMITE, NIX
Pyrethrins RID

AGENTS FOR PIGMENTATION DISORDERS

Hydroquinone VARIOUS
Methoxsalen OXSORALEN

AGENTS FOR PSORIASIS

Acitretin SORIATANE
Calcipotriene DOVONEX, SORILUX
Calcitriol VECTICAL
Tazarotene TAZORAC

KERATOLYTIC AGENTS

Coal tar OXIPOR, SCYTERA
Salicylic acid SALEX

TRICHOGENIC AGENTS

Finasteride PROPECIA
Minoxidil ROGAINE